Fraud & Anomaly Detection

Cause-first risk framework using real G'Contest banking data

1,418,030	35,451	8,662
Transactions scored	High/Critical cases	Customers impacted

- No synthetic data and no fake fraud labels.
- Output is a review queue with explainable risk reasons.
- Framework aligns with risk-based banking controls: step-up authentication, manual review, AML escalation.

Why Cause-First

BGK note: identify root causes before introducing the model

- Account takeover / identity compromise: new device, new IP, night access, new beneficiary, late-stage digital activity.
- Unauthorized transfer / capital outflow: outside-bank transfer, amount above customer baseline, daily burst, cash-out versus CASA.
- AML network / mule-account pattern: shared IP/device/beneficiary, round high-value transfers, repeated external transfers.

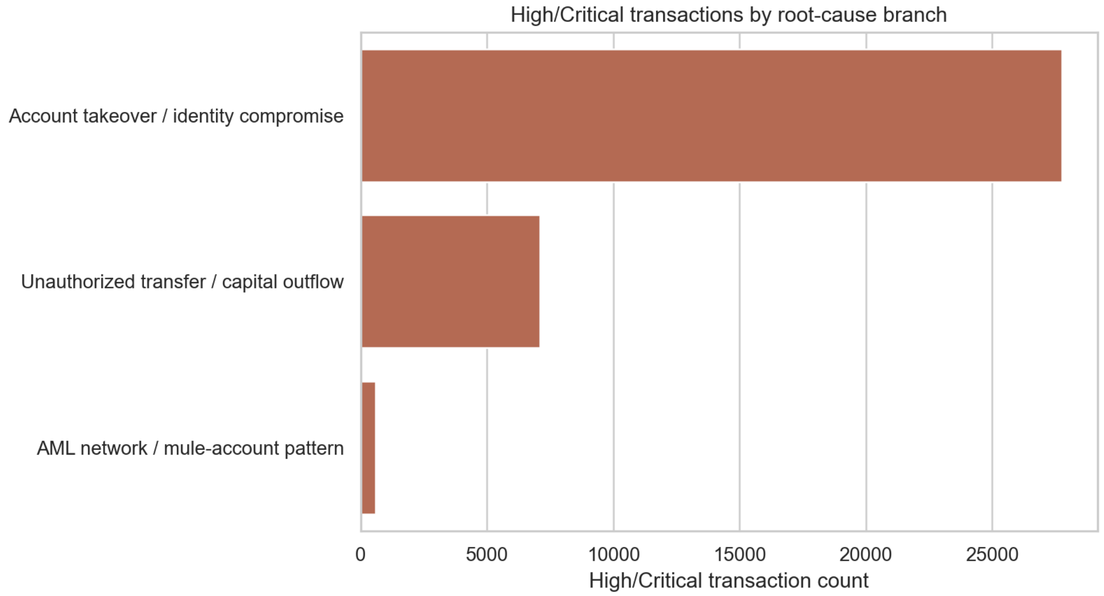
Framework

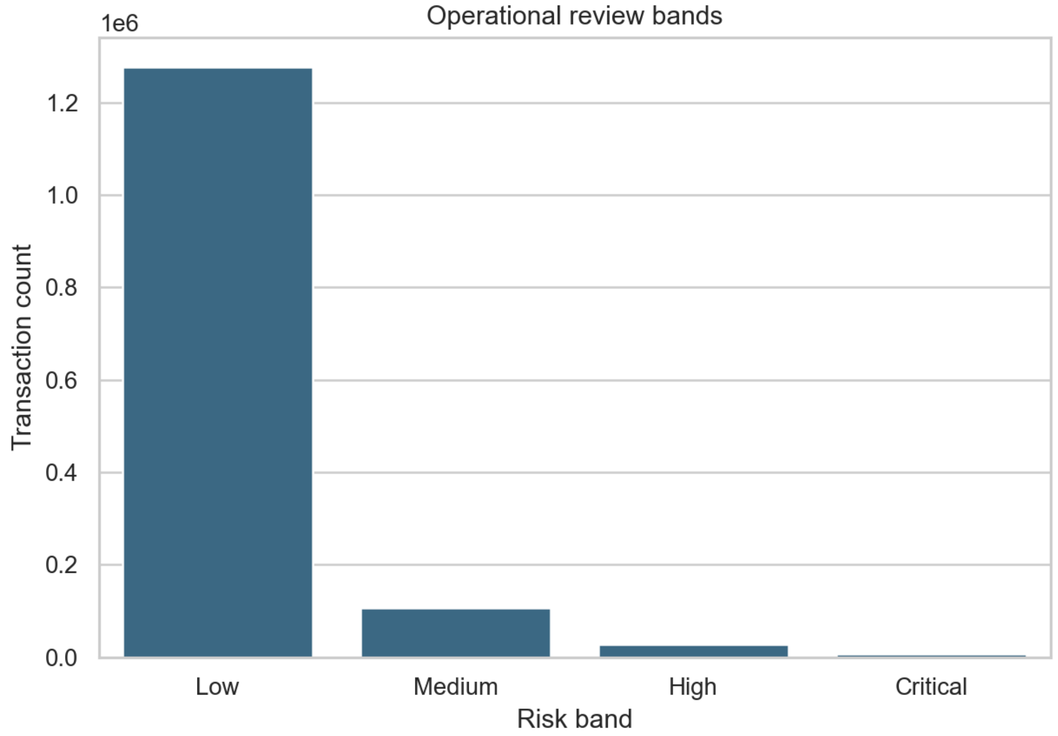
Data dictionary -> behavioral baseline -> root-cause rules -> anomaly model -> xAI review queue

1. Data	Customer, Transaction, Activity, Deposit, Lending, Card
2. Baseline	Customer amount P95, daily frequency, known device/IP/beneficiary
3. Root causes	ATO, unauthorized transfer, AML network scores
4. Model	Isolation Forest catches unusual combinations without fraud labels
5. xAI	Reason codes, SHAP surrogate, recommended action

Key Insights

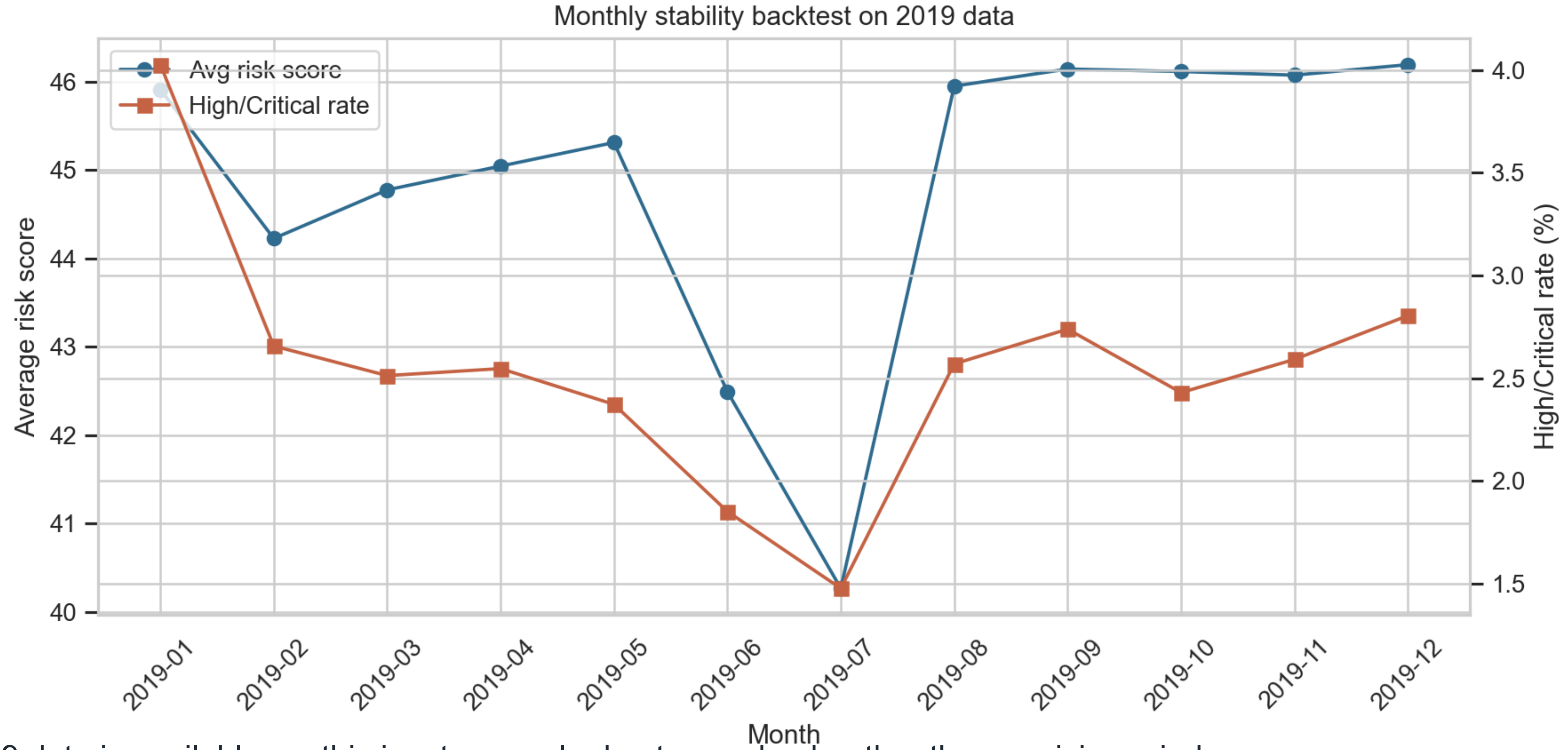
Top-risk queue is explainable by root-cause branch





Temporal Stability

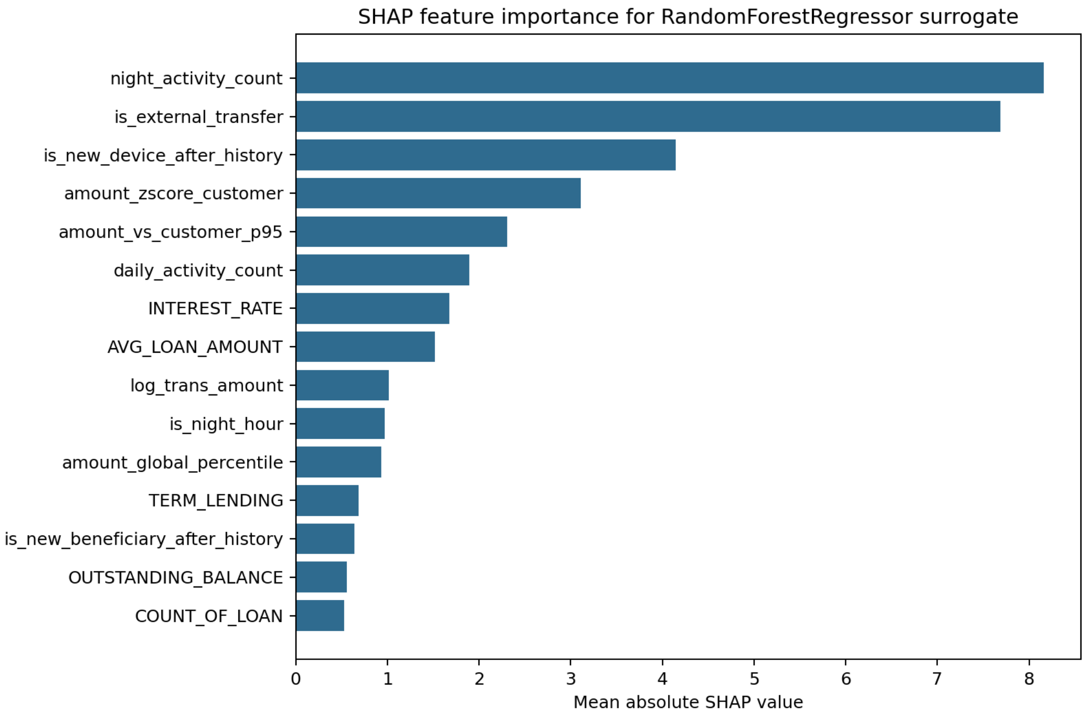
2019 monthly/quarterly backtest for review-rate stability

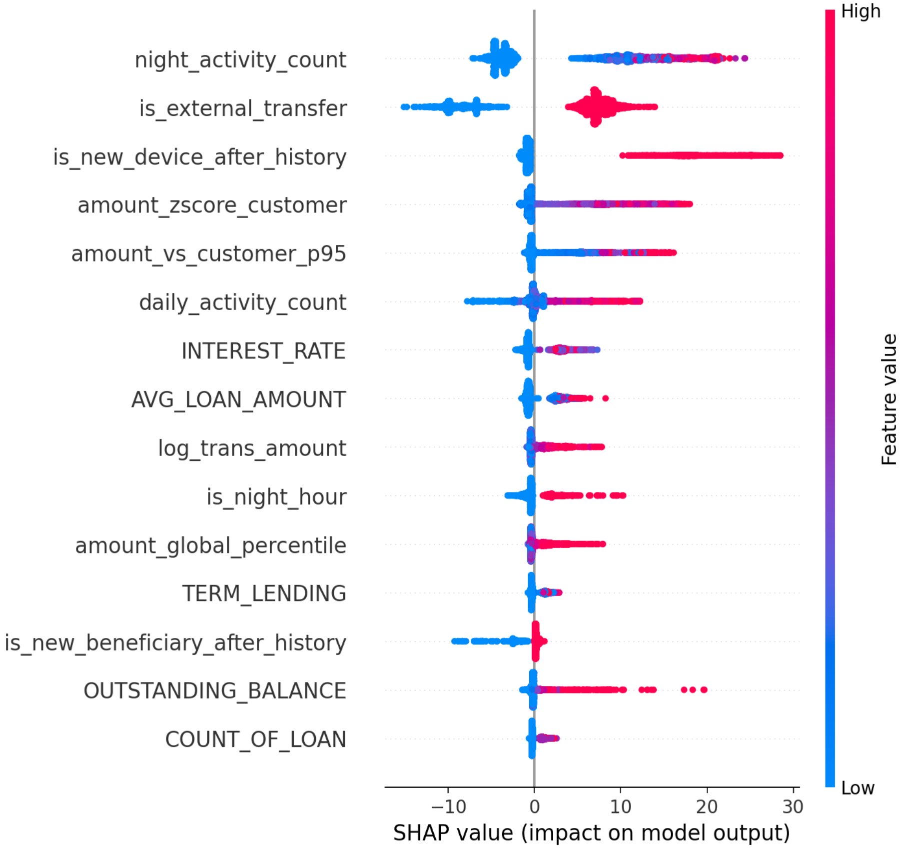


- Only 2019 data is available, so this is a temporal robustness check rather than a crisis-period validation.
- The monitoring KPI is not accuracy without labels; it is review queue volume, high-risk rate, and reason-code consistency over time.

xAI Engine

SHAP explains a tree surrogate of the hybrid risk score





Operational Recommendations

Turn model output into bank actions

- Critical: temporary hold, near-real-time manual review, customer verification, step-up authentication.
- High: same-day review queue and device/IP/beneficiary investigation.
- AML branch: network escalation rather than single-transaction review only.
- Medium: enhanced monitoring; escalate if repeated within the next review window.
- Feedback loop: investigator decisions become labels for Precision@K, Recall@K and supervised model tuning.

Live Demo

Streamlit advisor for transaction/customer review

- Run `streamlit run src/demo_app.py`.
- Select a top-risk transaction or enter a CUSTOMER_NUMBER.
- The advisor returns risk band, root-cause branch, reason codes, recommended action and SHAP evidence.
- This is designed for a short judging demo or screen-recorded walkthrough.